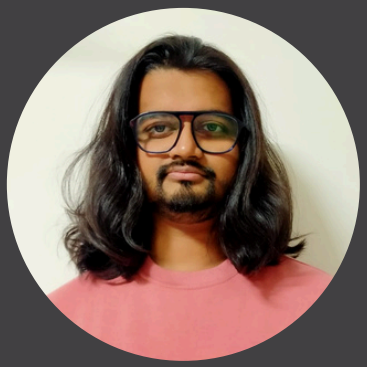


Utkarsh Shrivastav

Senior AI/ML Engineer

CONTACT

Address
Bengaluru, IN
"Open to relocation / remote"

Phone
+91 9717274326

Web
utkarsh1711sh@gmail.com
https://www.linkedin.com/in/utkarsh-shrivastav-17nov/

SKILLS

- **Core AI / ML**
  - Machine Learning, Deep Learning
  - Computer Vision, NLP
  - CNNs, Transformers, Siamese Networks
  - Supervised, Unsupervised & Metric Learning
  - Data Analysis
- **Generative AI & LLMs**
  - LangChain, Vector Databases
  - RAG Pipelines, Agent-based Systems.
  - Prompt Engineering.
  - LLMs: GPT-style models.
- **Deep Learning Frameworks**
  - PyTorch, TensorFlow, Keras
  - Custom Model Architectures
  - Model Training, Evaluation & Optimization
- **ML-Ops, Deployment & Systems**
  - End-to-End ML Pipelines (Training → Deployment → Monitoring)
  - Model Deployment (Cloud & Edge)
  - Docker, Git
  - Performance Optimization & Inference Tuning
  - CI/CD for ML Pipelines
  - FastAPI, Sheldon
- **Cloud Platforms**
  - AWS, GCP, Microsoft Azure
- **Programming**
  - Python

PROFILE

Senior AI/ML Engineer with 6+ years of experience designing, building, and deploying production-grade **Computer Vision, Deep Learning, and Generative AI** systems across cloud and edge environments. Strong expertise in **CNNs, Transformers, LLM-based RAG pipelines, and end-to-end ML lifecycle (data → training → deployment → monitoring).**
Proven track record of delivering real-world AI solutions in **waste management, anti-counterfeiting, and intelligent automation**, including live deployments for global brands such as **Nestlé (Maggi Masala)** and **Harpic**. Experienced in model optimization, scalable system design, and cross-functional collaboration to translate complex business problems into robust AI products.

EXPERIENCE

May-2023 to Present
Ficha

SENIOR AI ENGINEER

- Designed, trained, and deployed production-grade computer vision and deep learning models using TensorFlow and PyTorch, supporting scalable cloud and edge inference pipelines.
- Architected end-to-end ML pipelines covering data preprocessing, model training, evaluation, deployment, and performance monitoring in production environments.
- Led model architecture selection and experimentation by evaluating state-of-the art computer vision approaches to identify optimal solutions for real-world use cases.
- Optimized inference latency and runtime performance for edge deployments through algorithmic improvements and system-level optimizations.
- Designed and built supporting infrastructure to enable efficient training, deployment, and monitoring of machine learning models.
- Translated business requirements into scalable AI-driven solutions using predictive modeling and advanced statistical techniques.
- Processed and analyzed large-scale structured and unstructured datasets to extract actionable insights using machine learning and data mining methods.
- Owned the full project lifecycle end-to-end, from research and prototyping to production deployment and documentation.
- Led and contributed to computer vision initiatives involving image augmentation, filtering, and data quality improvement for robust model performance.
- Developed an internal Python package to standardize and accelerate deep learning model training, evaluation, and testing workflows.
- Conducted rigorous validation and testing to ensure reliability, accuracy, and maintainability of deployed models.
- Collaborated closely with cross-functional teams (product, engineering, operations) to ensure timely delivery and alignment with business goals.
- Achieved a 3.5–5× inference speedup and significantly reduced deployment turnaround time through end-to-end pipeline and optimization improvements.

Feb-2020 to April-2023
Ennoventure Technologies Pvt. Ltd, Bengaluru

ML ENGINEER

- Developed and optimized machine learning models using TensorFlow and PyTorch, achieving enhanced accuracy and runtime efficiency.
- Conceptualized scalable business solutions through predictive modeling and advanced statistics, contributing to the successful management of live products for Maggi and Harpic.
- Specialized in computer vision projects, focusing on image augmentation, filtering, and efficient data processing.
- Independently managed project lifecycles, documenting scalable solutions and collaborating for timely completion.
- Designed and implemented a Python package/tool streamlining deep learning model training and testing, ensuring accuracy and ease of use.

EDUCATION

2015-2019
BE - COMPUTER SCIENCE
CDGI, Indore, M.P, IN

CERTIFICATIONS

- **Introduction to TensorFlow for AI, ML, and DL** | Coursera | June 2021
- **Convolutional Neural Networks in TensorFlow** | Coursera | June 2021
- **Natural Language Processing in TensorFlow** | Coursera | June 2021
- **Sequences. Time Series and Prediction** | Coursera | June 2021
- **App Development (Windows platform using C#)** | Microsoft Training Program| September 2015
- **Web Development(using Python)| Microsoft Training Program**| April 2016

HOBBIES



PROJECTS (FICHA)

COCON

The COCON is a connected, AI-powered container placed on the sorting bins within trash rooms in collective housing. It helps raise awareness of the correct sorting action by reducing sorting errors. Thanks to its artificial intelligence, it helps improve waste management efficiency and cleanliness of trash rooms and make sorting interactive and fun.

- Led research and development to evaluate and select optimal computer vision model architectures for object characterization and sorting error detection.
- Designed and trained task-specific deep learning models tailored to real-world waste classification and filtering scenarios.
- Deployed production-grade models across both cloud and edge environments, enabling scalable and low-latency inference in on-site installations.
- Optimized model performance and inference reliability through extensive testing, validation, and iterative refinement under real-world conditions.
- Collaborated with cross-functional teams (product, hardware, and engineering) to address system-level challenges and ensure seamless end-to-end integration.

CAMIA

CAMIA is an AI-powered vision system installed in dump trucks to characterize the sorting quality of each bin deposited and detect sorting errors. Equipped with an artificial intelligence system, it allows communities and collection operators to reduce sorting refusals on their territory.

- Led research and development to identify and benchmark optimal computer vision model architectures for waste characterization and sorting quality assessment.
- Designed and trained deep learning models to detect sorting errors and evaluate bin content under diverse environmental and operational conditions.
- Deployed production-ready models across cloud and edge environments, supporting low-latency inference directly on collection vehicles.
- Optimized model robustness and performance through extensive real-world testing, validation, and iterative improvement.
- Collaborated with cross-functional teams (hardware, data, and operations) to integrate AI models into end-to-end collection workflows and resolve system-level challenges.
- Addressed challenges related to motion, lighting variation, and constrained compute environments in mobile edge deployments.

LLM Chatbot

Developed an advanced chatbot using LangChain, Vector DB, and an agent framework to enhance user interaction and query resolution. LangChain was utilized for managing complex dialogue flows, Vector DB for efficient data retrieval, and the agent framework for dynamic response generation. This integration resulted in a chat bot capable of providing coherent, contextually relevant, and rapid responses to user inquiries.

- Utilized LangChain for handling complex dialogue and maintaining conversation context.
- Implemented Vector DB for fast and accurate data retrieval.
- Integrated an agent framework to enable adaptive and dynamic responses.

PROJECTS (ENNOVENTURE)

Invisible Cryptographic Signature (Maggi Masala & Harpic)

- Served as lead ML engineer for live anti-counterfeit systems leveraging deep learning-based image preprocessing, classification, and filtering.
- Integrated custom ML models into production systems used by global FMCG brands.

Image Classification & Filtering (CNN-based Systems)

- Designed and deployed CNN-based classification and filtering pipelines for real world image data across web and cloud platforms.
- Focused on model robustness, performance optimization, and deployment reliability.

Object Detection Systems (YOLO, SSD)

- Developed and deployed object detection models using YOLO and SSD architectures to identify regions of interest in real-world images.
- Optimized detection accuracy and inference performance for cloud-based applications.

Synthetic Image Generation & Augmentation

- Built computer vision pipelines for synthetic image generation by blending digital objects with real-world backgrounds.
- Applied advanced augmentation, enhancement, and realism techniques to improve training data quality and model generalization.

One-Shot Learning

One-Shot Learning & Similarity Modeling

- Led development of a one-shot learning system using Siamese Neural Networks to measure visual similarity between images.
- Designed models suitable for real-time inference and web-based deployment.